HW-5

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The database was reset with `hw5.sql` before running these tests, in the order shown. Menu options are omitted from the output.

List countries (starter code)

all countries after resetting the database with `hw5.sql`:

```
Enter your choice (1-8): 1

Belgium (BE), per capita gdp $48700, inflation rate 2.9%
Canada (CA), per capita gdp $43100, inflation rate 2.4%
Germany (DE), per capita gdp $52100, inflation rate 2.2%
Iceland (IS), per capita gdp $52100, inflation rate 3.1%
Mexico (MX), per capita gdp $15300, inflation rate 4.7%
Netherlands (NL), per capita gdp $57000, inflation rate 3.3%
United States of America (US), per capita gdp $46900, inflation rate 3.8%
```

Show country

existing country:

```
Enter your choice (1-8): 2

Country code: CA

Canada (CA), per capita gdp $43100, inflation rate 2.4%
```

missing country:

```
Enter your choice (1-8): 2

Country code: XX

No matching country.
```

Add country

new country:

```
Enter your choice (1-8): 3

Country code: NZ
Country name: New Zealand
Country gdp: 42000
Country inflation: 2.0

Country added.
```

same code again with different values:

```
Enter your choice (1-8): 3

Country code: NZ
Country name: New Zealand Revised
Country gdp: 50000
Country inflation: 3.0

Country already exists.
```

verify: NZ kept its original name, gdp, and inflation:

```
Enter your choice (1-8): 2

Country code: NZ

New Zealand (NZ), per capita gdp $42000, inflation rate 2.0%
```

Update country

existing country:

```
Enter your choice (1-8): 4

Country code: CA
Country gdp: 46000
Country inflation: 2.0

Country updated.
```

verify: CA has the new gdp and inflation:

```
Enter your choice (1-8): 2

Country code: CA

Canada (CA), per capita gdp $46000, inflation rate 2.0%
```

same values again still reports updated:

```
Enter your choice (1-8): 4

Country code: CA
Country gdp: 46000
Country inflation: 2.0

Country updated.
```

missing country:

```
Enter your choice (1-8): 4

Country code: AB
Country gdp: 46000
Country inflation: 2.0

Country does not exist.
```

Remove border

border stored as (US, CA), entered in reverse order:

```
Enter your choice (1-8): 5

Country 1 code: CA
Country 2 code: US

Border removed.
```

verify: repeating it finds no border:

```
Enter your choice (1-8): 5

Country 1 code: CA
Country 2 code: US

No border exists between the countries.
```

border entered in stored order (BE, NL):

```
Enter your choice (1-8): 5

Country 1 code: BE
Country 2 code: NL

Border removed.
```

countries that never shared a border:

```
Enter your choice (1-8): 5

Country 1 code: US
Country 2 code: DE

No border exists between the countries.
```

Add province with city

new province and city:

```
Enter your choice (1-8): 6

Country code: CA
```

```
Province name: Alberta
Province area: 255541
City name: Calgary
City population: 1678073

Province and city added.
```

existing province: no city added, area unchanged:

```
Enter your choice (1-8): 6

Country code: CA
Province name: Alberta
Province area: 661848
City name: Edmonton
City population: 600000

Province already exists.
```

verify: Alberta has only Calgary (Edmonton was not added):

```
Enter your choice (1-8): 7

Country code (enter to skip): CA
Province name (enter to skip): Alberta
Minimum population (enter to skip):
Maximum population (enter to skip):

Calgary, Alberta (CA), population 1678073

See more rows (y/n): n
```

missing country: the province insert fails (foreign key):

```
Enter your choice (1-8): 6

Country code: ZZ
Province name: Nowhere
Province area: 1000
City name: Ghost Town
City population: 10
Database error: insert or update on table "province" violates foreign key constraint "province_country_code_fkey"
DETAIL:  Key (country_code)=(ZZ) is not present in table "country".
```

province insert succeeds, then the city insert fails (name longer than 40 characters):

```
Enter your choice (1-8): 6

Country code: CA
Province name: Yukon
Province area: 482443
City name: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
City population: 30000
Database error: value too long for type character varying(40)
```

verify: the rollback also undid the Yukon province insert:

```
Enter your choice (1-8): 7

Country code (enter to skip): CA
Province name (enter to skip): Yukon
```

```
Minimum population (enter to skip):
Maximum population (enter to skip):

No matching cities.
```

Search cities

country filter only, more than five rows: display a second batch:

```
Enter your choice (1-8): 7

Country code (enter to skip): US
Province name (enter to skip):
Minimum population (enter to skip):
Maximum population (enter to skip):

Seattle, Washington (US), population 737015
Portland, Oregon (US), population 652503
Spokane, Washington (US), population 228989
Vancouver, Washington (US), population 190915
Eugene, Oregon (US), population 176654

See more rows (y/n): y

Salem, Oregon (US), population 175535

See more rows (y/n): n
```

no filters (all cities): stop after the first batch:

```
Enter your choice (1-8): 7

Country code (enter to skip):
Province name (enter to skip):
Minimum population (enter to skip):
Maximum population (enter to skip):

Toronto, Ontario (CA), population 2794356
Calgary, Alberta (CA), population 1678073
Munich, Bavaria (DE), population 1512491
Zapopan, Jalisco (MX), population 1476491
Guadalajara, Jalisco (MX), population 1385629

See more rows (y/n): n
```

province name in two countries (Limburg), equal min and max: ties sorted by country code, then city name:

```
Enter your choice (1-8): 7

Country code (enter to skip):
Province name (enter to skip): Limburg
Minimum population (enter to skip): 100000
Maximum population (enter to skip): 100000
```

```
Genk, Limburg (BE), population 100000
Sittard, Limburg (NL), population 100000
Venlo, Limburg (NL), population 100000

See more rows (y/n): n
```

population bounds only: cities at exactly 12000 and 15000 are included:

```
Enter your choice (1-8): 7

Country code (enter to skip):
Province name (enter to skip):
Minimum population (enter to skip): 12000
Maximum population (enter to skip): 15000

Portland, Ontario (CA), population 15000
Neustadt, Bavaria (DE), population 13000
Neustadt, Saxony (DE), population 12000

See more rows (y/n): n
```

country and minimum population:

```
Enter your choice (1-8): 7

Country code (enter to skip): DE
Province name (enter to skip):
Minimum population (enter to skip): 600000
Maximum population (enter to skip):

Munich, Bavaria (DE), population 1512491
Leipzig, Saxony (DE), population 616093

See more rows (y/n): n
```

no matching cities:

```
Enter your choice (1-8): 7

Country code (enter to skip): CA
Province name (enter to skip): FooBar
Minimum population (enter to skip):
Maximum population (enter to skip):

No matching cities.
```